Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 5 — Sequences and series
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第5章 · 数列与级数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

An arithmetic sequence has fifth term 17 and twelfth term 45.

(a) Find its first term and common difference. (3)

(b) Find the sum of its first 20 terms. (2)

(Total for Question 1 is 5 marks)

A geometric sequence has first term 12 and common ratio $-1/2$.

(a) Find its fourth and fifth terms. (2)

(b) Find the sum to infinity, explaining why it exists. (3)

(Total for Question 2 is 5 marks)

3.

A theatre has 16 rows. The first row has 18 seats, and each later row has 2 more seats than the preceding row.

(a) Find the number of seats in the final row and the total number of seats. (4)

(b) Every seat in the first 6 rows costs £20, and all remaining seats cost £15. Find

(i) the number of £20 seats, (2)

(ii) the total ticket income when all seats are sold. (3)

(Total for Question 3 is 9 marks)

4.

The first three terms of an arithmetic sequence are k, $2k - 3$ and $4k - 11$.

(a) Find k. (2)

(b) Find the least n for which the nth term exceeds 100. (3)

(Total for Question 4 is 5 marks)

5.

A geometric sequence has second term 6 and fifth term 162. Its common ratio is positive.

(a) Find its first term and common ratio. (3)

(b) Find the sum of the first 6 terms. (2)

(Total for Question 5 is 5 marks)

6. P2-C05-R06 · P2 · Chapter 5

The sum of the first n terms of an arithmetic sequence is $S_n = 3n^2 + 2n$.

(a) Find its first term and common difference. (3)

(b) Find the least positive integer n such that $S_n > 500$. (3)

(Total for Question 6 is 6 marks)

7.

A bouncing ball is dropped from 5 m. Each rebound reaches $3/5$ of the preceding maximum height.

(a) Find the height reached on the fourth rebound. (2)

(b) Find the total vertical distance travelled from its initial release, including the first fall, over infinitely many rebounds under the model. (3)

(Total for Question 7 is 5 marks)

An arithmetic sequence has $a = 40$ and $d = -3$.

(a) Find the last positive term and its position. (3)

(b) Find the sum of all positive terms. (2)

(Total for Question 8 is 5 marks)

9.

The first term of a geometric series is 8 and its sum to infinity is 20.

(a) Find the common ratio. (2)

(b) Find the sum of the first 4 terms and the exact sum of the remaining terms. (4)

(Total for Question 9 is 6 marks)

A sequence is defined by $u_1 = 4$ and $u_{n+1} = 2u_n - 3$.

(a) Find u_2, u_3, u_4. (2)

(b) Let $v_n = u_n - 3$. Show that (v_n) is geometric and find u_n in terms of n. (4)

(Total for Question 10 is 6 marks)

11.

A runner increases the daily distance by 0.5 km each day, starting at 3 km on day 1.

(a) Find the distance on day 12 and the total distance over the first 12 days. (4)

(b) Find the first day on which the cumulative distance exceeds 100 km. (3)

(Total for Question 11 is 7 marks)

For $r > 0$, the numbers $r - 1$, $r + 1$, $2r + 2$ are consecutive terms of a geometric sequence.

(a) Find r. (3)

(b) Find the sum of the first 5 terms when these are the first three terms. (3)

(Total for Question 12 is 6 marks)

$$S = \sum_{k=1}^{20} (3k - 2).$$

(a) Find S. (3)

(b) Find $\sum_{k=6}^{20} (3k - 2)$. (3)

(Total for Question 13 is 6 marks)

The sum of the first 10 terms of an arithmetic sequence is 180, and the sum of the first 20 terms is 660

.

(a) Find the first term and common difference. (3)

(b) Find the sum of terms 11 to 20 inclusive. (2)

(Total for Question 14 is 5 marks)

A geometric series has positive first term a and common ratio r with $0 < r < 1$. The sum of its first two terms is 18 and its sum to infinity is 24.

(a) Find r. (4)

(b) Find a and the least n for which $S_n > 23$. (4)

(Total for Question 15 is 8 marks)

A charity receives £100 in week 1, then £15 more each week than in the preceding week.

(a) Find the total received in 8 weeks. (2)

(b) From week 9 onwards the weekly amount stays equal to the week 8 amount. Find

(i) the fixed weekly amount, (2)

(ii) the first week by the end of which the cumulative total exceeds £2000. (3)

(Total for Question 16 is 7 marks)

The first three terms of a sequence are $2, 5, 8$.

(a) Assuming the sequence is arithmetic, find its twentieth term. (2)

(b) The formula $u_n = 3n - 1 + (n-1)(n-2)(n-3)$ also gives these first three terms. Find u_4 and explain why three listed terms alone do not prove that a sequence is arithmetic. (4)

(Total for Question 17 is 6 marks)

18.

A geometric sequence has first term 81 and common ratio $2/3$.

(a) Find its fifth term and the sum of its first five terms. (4)

(b) Find the sum to infinity of the sequence formed by its even-numbered terms. (4)

(Total for Question 18 is 8 marks)

19.

An arithmetic sequence has first term a and common difference $d \neq 0$. Its fourth, seventh and thirteenth terms form a geometric sequence in that order.

(a) Show that $a = 0$. (4)

(b) Given that the fourth term is 6, find d and the common ratio of these three selected terms. (3)

(Total for Question 19 is 7 marks)

An account receives £200 at the beginning of each year. At the end of each year its balance is multiplied by 1.05. There are no other transactions.

(a) Find the balance at the end of year 3. (3)

(b) Write a formula for the balance at the end of year n. (3)

(Total for Question 20 is 6 marks)

$u_n = 7 - 2n$ for positive integers n.

(a) Find the first negative term and its position. (2)

(b) Find $\sum_{n=4}^{12} u_n$. (3)

(Total for Question 21 is 5 marks)

A geometric series has first term 5 and ratio r, with $|r| < 1$. Its sum to infinity is $10/3$.

(a) Find r. (2)

(b) Find the sum of its first 6 terms and state whether it is above or below the infinite sum. (4)

(Total for Question 22 is 6 marks)

A sequence is defined by $u_1 = 2$ and $u_{n+1} = u_n + 3n$.

(a) Find u_2, u_3, u_4. (2)

(b) By summing the increments, show that $u_n = 2 + \frac{3}{2}n(n-1)$. (3)

(Total for Question 23 is 5 marks)

The third term of a geometric sequence is 12 and the fifth is 48.

(a) Find all possible real common ratios and corresponding first terms. (3)

(b) For which of these sequences is the sum of the first four terms negative? Find that sum. (3)

(Total for Question 24 is 6 marks)

The first n terms of an arithmetic sequence have sum $n(2n + 3)$.

(a) Find a formula for the nth term. (3)

(b) Find the sum of terms 21 to 40 inclusive. (3)

(Total for Question 25 is 6 marks)

A machine produces 1200 items in its first hour. Each subsequent hour it produces 90% as many as in the previous hour. Production is modelled without rounding the output in individual hours.

(a) Find the total production in the first 8 hours, to the nearest item. (3)

(b) Find the greatest total production predicted over infinitely many hours and explain why it is finite. (3)

(Total for Question 26 is 6 marks)

The sum of the first n terms of a sequence is $S_n = 2n^2 + 5n$, for positive integers n.

(a) Show that the nth term is $u_n = 4n + 3$. (2)

(b) The sum of the terms from u_{n+1} to u_{2n} inclusive is 650.

(i) Show that $n(6n + 5) = 650$. (2)

(ii) Find n and the value of u_{2n}. (2)

(Total for Question 27 is 6 marks)

A workshop produces 80 parts in its first week. Under plan A, production increases by 12 parts each subsequent week. Under plan B, production increases by 10% each subsequent week. Production under plan B is modelled without rounding to whole parts.

(a) Find the total production over the first 8 weeks under plan A. (2)

(b) For plan B,

(i) find the total production over the first 8 weeks to the nearest whole part. (2)

(ii) find the first week in which weekly production exceeds 200 parts. (2)

(Total for Question 28 is 6 marks)

A geometric sequence has positive terms. Its second term is 18 and its fifth term is $243/4$.

(a) Find the first term a and common ratio r. (2)

(b) The first n terms have sum S_n.

(i) Show that $S_{n+1} = \frac{3}{2}S_n + 12$. (3)

(ii) Find the least n for which $S_n > 1000$. (2)

(Total for Question 29 is 7 marks)

At the end of each year, a trust pays a grant. The first grant is £5000, and each subsequent grant is 4% less than the preceding grant. No other payments are made.

(a) Find the total of all grants over the first 12 years, to the nearest pound. (2)

(b) The trust has £125000 available for these grants.

(i) Show that the total value of all grants in the infinite model is £125000. (3)

(ii) Find the least number of complete years after which the money not yet paid is less than £50000. (3)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C05-R01 · 5 marks

(a) · 3 marks

$a=1, d=4$.

a+4d=17，a+11d=45，相减7d=28。

M1	两项条件
M1	消元
A1	a与d

(b) · 2 marks

780.

S20=20[2+19×4]/2=780。

M1	正确等差和式
A1	总和

Question 2

P2-C05-R02 · 5 marks

(a) · 2 marks

$-3/2$ and $3/4$.

第n项12(−1/2)^{n−1}，注意符号交替。

M1	正确项指数
A1	两项

(b) · 3 marks

8; $|r| = 1/2 < 1$.

无穷和12/[1−(−1/2)]=8，负公比仍可收敛。

B1	收敛条件
M1	无穷和公式
A1	和

Question 3

P2-C05-R03 · 9 marks

(a) · 4 marks

Final row 48; total 528.

a16=18+15×2；S16=16(18+48)/2。

M1	最后一行项数
A1	48
M1	等差求和
A1	528

(b) Every seat in the first 6 rows costs £20, and all remaining seats cost £15. Find

(i) · 2 marks

138.

S6=6[36+5×2]/2=138。

M1	前六行等差和
A1	席位

(ii) · 3 marks

£8610.

20×138+15×(528−138)=2760+5850。

M1	两票价分组
M1	后组席位
A1	总收入

Question 4

P2-C05-R04 · 5 marks

(a) · 2 marks

$k = 5$.

两差相等k−3=2k−8，得5。

M1	等差条件
A1	k

(b) · 3 marks

$n = 49$.

a=5,d=2，5+2(n−1)>100给n>48.5，最小49。

M1	正确通项
M1	严格不等式与整数
A1	n

Question 5

P2-C05-R05 · 5 marks

(a) · 3 marks

$a = 2, r = 3.$

ar=6，ar^4=162，除得r^3=27取3，a=2。

M1	项比消元
M1	求公比
A1	a与r

(b) · 2 marks

$728.$

$2(3^6-1)/(3-1)=728$。

M1	正确有限等比和
A1	总和

Question 6

P2-C05-R06 · 6 marks

(a) · 3 marks

$a = 5, d = 6.$

a=S1=5；a2=S2−S1=16−5=11，差6；或比较和式系数。

M1	从和求第一项
M1	求第二项或比较系数
A1	a与d

(b) · 3 marks

$n = 13.$

$3n^2+2n-500>0$，或检查S12=456，S13=533，且正n上S递增。

M1	正确阈值
M1	相邻整数验证及增长
A1	n

Question 7

P2-C05-R07 · 5 marks

(a) · 2 marks

$81/125 = 0.648$ m.

第k次反弹高5(3/5)^k，第四次5×81/625。

M1	正确反弹索引
A1	高度

(b) · 3 marks

20 m.

初降5，加两倍所有反弹高度和：5+2[3/(1−3/5)]=20。

M1	初次下落单独计入
M1	上下双程等比和
A1	路程

Question 8

P2-C05-R08 · 5 marks

(a) · 3 marks

Last positive term $a_{14} = 1$.

40−3(n−1)>0给n<43/3，最大整数14，代入1。

M1	正通项条件
M1	最大整数索引
A1	位置和项

(b) · 2 marks

287.

S14=14(40+1)/2=287。

M1	使用正确末项及项数
A1	总和

Question 9

P2-C05-R09 · 6 marks

(a) · 2 marks

$r = 3/5$.

8/(1−r)=20给1−r=2/5；结果满足绝对值小于1。

M1	无穷和关系
A1	r

(b) · 4 marks

$S_4 = 2176/125$; remaining sum $324/125$.

S4=20[1−(3/5)⁴]=2176/125；用20减S4得324/125。

M1	有限等比和
A1	S4
M1	从无穷和相减
A1	余和

Question 10

P2-C05-R10 · 6 marks

(a) · 2 marks

$5, 7, 11$.

按递推逐次计算，下一项依赖上一项而非索引n。

M1	正确逐项递推
A1	三项

(b) · 4 marks

$v_{n+1} = 2v_n$, $v_1 = 1$; $u_n = 3 + 2^{n-1}$.

两边减3得u_{n+1}−3=2(u_n−3)，所以v为公比2等比，再还原u。

M1	正确改写递推
M1	确定首项和公比
M1	还原u
A1	通项

Question 11

P2-C05-R11 · 7 marks

(a) · 4 marks

8.5 km; 69 km.

$a_{12}=3+11\times.5$，$S_{12}=12(3+8.5)/2$。

M1	通项索引
A1	第12天
M1	求和
A1	总距离

(b) · 3 marks

Day 16.

$S_n=n[6+.5(n-1)]/2=.25n^2+2.75n$；$S_{15}=97.5$，$S_{16}=108$。

M1	累计和表达式
M1	检查相邻整数
A1	日数

Question 12

P2-C05-R12 · 6 marks

(a) · 3 marks

$r = 3$.

$(r+1)^2=(r-1)(2r+2)$，化为$(r+1)(r-3)=0$；$r>0$排除-1。

M1	等比中项平方条件
M1	解并筛选
A1	r

(b) · 3 marks

62.

首项2、公比2，$S_5=2(2^5-1)=62$。

M1	正确a与公比
M1	有限和
A1	总和

Question 13

P2-C05-R13 · 6 marks

(a) · 3 marks

590.

所加项为1、4、…、58，20项等差和20(1+58)/2。

M1	正确首末项与项数
M1	等差和
A1	S

(b) · 3 marks

555.

先五项和为5(1+13)/2=35，从590减去35。

M1	正确去除前五项
M1	首五和
A1	结果

Question 14

P2-C05-R14 · 5 marks

(a) · 3 marks

$a = 9/2, d = 3$.

2a+9d=36，2a+19d=66，相减10d=30，a=4.5。

M1	两个和条件
M1	消元
A1	a与d

(b) · 2 marks

480.

该分段为S20−S10=660−180。

M1	正确部分和差
A1	结果

Question 15

P2-C05-R15 · 8 marks

(a) · 4 marks

$r = 1/2$.

a(1+r)=18与a/(1−r)=24相除得1−r²=3/4，r²=1/4取正。

M1	两和条件
M1	消去a
M1	用正公比条件
A1	r

(b) · 4 marks

$a = 12$, least $n = 5$.

Sn=24(1−2^{−n})>23给2^n>24，n最小5。

M1	a
M1	正确有限和不等式
M1	整数比较
A1	n

Question 16

P2-C05-R16 · 7 marks

(a) · 2 marks

£1220.

S8=8[200+7×15]/2。

M1	等差求和式
A1	总额

(b) From week 9 onwards the weekly amount stays equal to the week 8 amount. Find

(i) · 2 marks

£205.

a8=100+7×15。

M1	正确索引
A1	金额

(ii) · 3 marks

Week 12.

额外k周需1220+205k>2000，k>780/205≈3.805，取4，再加8。

M1	正确分段累计不等式
M1	取额外整数周
A1	总周数

Question 17

P2-C05-R17 · 6 marks

(a) · 2 marks

59.

a=2,d=3，a20=2+19×3。

M1	识别a,d
A1	结果

(b) · 4 marks

$u_4 = 17$; an alternative rule matches the first three terms but not the fourth arithmetic term 11.

代4得12−1+6=17；共同前三项不足唯一规定后续规则。

M1	正确代n=4
A1	u4
M1	比较不同后续
A1	说明前提作用

Question 18

P2-C05-R18 · 8 marks

(a) · 4 marks

16; 211.

$a_5=81(2/3)^4=16$；$S_5=81[1-(2/3)^5]/(1/3)=211$。

M1	正确通项
A1	第五项
M1	有限和
A1	总和

(b) · 4 marks

$486/5$.

偶数项首项54，新公比$(2/3)^2=4/9$，无穷和$54/(1-4/9)$。

M1	正确新首项
M1	正确新公比
M1	无穷和
A1	精确结果

Question 19

P2-C05-R19 · 7 marks

(a) · 4 marks

$(a+6d)^2=(a+3d)(a+12d)$ gives $-3ad=0$, hence $a=0$.

两边展开，a^2和$36d^2$抵消，剩$12ad=15ad$；d非零才可推出a零。

M1	等比条件
M1	展开化简
M1	使用d非零
A1	a

(b) · 3 marks

$d=2$; ratio 2.

首项0，第四项$3d=6$，三选项6、12、24。

M1	正确d
M1	求三选项
A1	比值

Question 20

P2-C05-R20 · 6 marks

(a) · 3 marks

£662.03 to the nearest penny.

三笔分别积累3、2、1年，余额$200(1.05^3+1.05^2+1.05)=662.025$。

M1	正确三笔计息指数
M1	相加
A1	金额精度

(b) · 3 marks

$B_n = 210\dfrac{1.05^n - 1}{0.05}$.

按计息年数排列为$210+210(1.05)+\cdots+210(1.05)^{n-1}$，共n项等比和。

M1	正确等比首项及项数
M1	有限和
A1	公式

Question 21

P2-C05-R21 · 5 marks

(a) · 2 marks

$u_4 = -1$.

7−2n<0给n>3.5，首个正整数4。

M1	负项条件
A1	项与索引

(b) · 3 marks

-81.

共有9项，首−1末−17，等差和9(−18)/2=−81。

M1	正确项数
M1	首末项和
A1	结果

Question 22

P2-C05-R22 · 6 marks

(a) · 2 marks

$r = -1/2$.

5/(1−r)=10/3给1−r=3/2。

M1	无穷和方程
A1	r

(b) · 4 marks

$S_6 = 105/32$; below $10/3$.

S6=(10/3)(1−1/64)=105/32；偶数幂余项正被减去。

M1	正确有限和
A1	精确值
M1	比较余项符号
A1	方向

Question 23

P2-C05-R23 · 5 marks

(a) · 2 marks

$5, 11, 20.$

依次加3、6、9，而不是每步固定加3。

M1	正确索引递推
A1	三项

(b) · 3 marks

$u_n = 2 + 3\sum_{k=1}^{n-1} k = 2 + 3n(n-1)/2.$

从u1到un需要n−1次增量，增量构成3、6、…、3(n−1)的等差和。

M1	正确增量项数
M1	等差求和
A1	公式

Question 24

P2-C05-R24 · 6 marks

(a) · 3 marks

$r = 2$ or $r = -2$; $a = 3$ in both cases.

两项相除$r^2=4$，保留两符号；$ar^2=12$给$a=3$。

M1	项比条件
M1	两符号
A1	a与公比

(b) · 3 marks

$r = -2$; $S_4 = -15$.

$r=-2$时$3-6+12-24=-15$；$r=2$所有项正。

M1	分析两公比符号
M1	负公比四项或和式
A1	公比与和

Question 25

P2-C05-R25 · 6 marks

(a) · 3 marks

$a_n = 4n + 1$.

Sn−S(n−1)=2n²+3n−[2(n−1)²+3(n−1)]=4n+1，n=1也一致。

M1	相邻和差
M1	展开化简
A1	通项

(b) · 3 marks

2460.

S40−S20=40×83−20×43=3320−860。

M1	正确部分和差
M1	代入
A1	总和

Question 26

P2-C05-R26 · 6 marks

(a) · 3 marks

6834 items.

S8=1200(1−.9⁸)/.1=6834.39348，最后舍入。

M1	正确等比和
M1	不提前舍入
A1	整数产量

(b) · 3 marks

12000 items; $|r| = 0.9 < 1$.

无穷等比和1200/(1−.9)，各小时产量趋零；只是模型极限。

M1	无穷和
A1	数值
B1	收敛理由

Question 27

P2-C05-R27 · 6 marks

(a) · 2 marks

$u_n = 4n + 3$.

对$n \geq 2$，$u_n = S_n - S_{n-1} = 2[n^2 - (n-1)^2] + 5 = 4n + 3$；另核对$n = 1$有$u_1 = S_1 = 7$，公式同样成立。

M1	相邻部分和作差
A1	化简并核对首项

(b) The sum of the terms from u_{n+1} to u_{2n} inclusive is 650.

(i) · 2 marks

$S_{2n} - S_n = 6n^2 + 5n = 650$.

此段有n项，和为$S_{2n} - S_n = (8n^2 + 10n) - (2n^2 + 5n) = 6n^2 + 5n$。

M1	正确上下限对应部分和
A1	所得方程

(ii) · 2 marks

$n = 10$; $u_{20} = 83$.

$6n^2 + 5n - 650 = (n - 10)(6n + 65) = 0$。由$n$为正整数取$10$，所以$u_{20} = 4(20) + 3 = 83$。

M1	解二次方程
A1	合法整数与末项

Question 28

P2-C05-R28 · 6 marks

(a) · 2 marks

976 parts.

等差数列首项80、公差12，$S_8 = 8[2(80) + 7(12)]/2 = 976$。

M1	等差和
A1	总数

(b) For plan B,

(i) · 2 marks

915 parts.

等比和$80(1.1^8 - 1)/(1.1 - 1) = 914.871\ldots$，最终取整为915。

M1	等比和
A1	正确最终取整

(ii) · 2 marks

Week 11.

$80(1.1)^{n-1} > 200$，得$n - 1 > \log 2.5 / \log 1.1 = 9.613\ldots$，故$n = 11$；这比较的是单周产量而非累计产量。

M1	对数解指数阈值
A1	首个周次

Question 29

P2-C05-R29 · 7 marks

(a) · 2 marks

$a = 12, r = 3/2$.

$ar = 18, ar^4 = 243/4$。相除$r^3 = 27/8$，所以$r = 3/2, a = 12$。

M1	两项条件相除
A1	a,r

(b) The first n terms have sum S_n.

(i) · 3 marks

$S_n = 24[(3/2)^n - 1]$, hence the relation.

有限等比和$S_n = 12[(3/2)^n - 1]/(1/2) = 24[(3/2)^n - 1]$；乘$3/2$再加$12$得到$24[(3/2)^{n+1} - 1]$。

M1	写出部分和
M1	对比n与n+1
A1	关系成立

(ii) · 2 marks

$n = 10$.

$24[(3/2)^n - 1] > 1000$即$(3/2)^n > 128/3$。取对数$n > 9.257067\ldots$，所以最小正整数为10。

M1	正确和的阈值
A1	最小整数

Question 30

P2-C05-R30 · 8 marks

(a) · 2 marks

£48411.

首项5000、公比0.96，前12项和$5000(1-0.96^{12})/0.04 = 48411.\ldots$，四舍五入到英镑。

M1	有限等比和
A1	金额

(b) The trust has £125000 available for these grants.

(i) · 3 marks

$5000/(1-0.96) = 125000$.

因为$|0.96| < 1$，无穷等比和存在，值为125000。

B1	收敛条件
M1	无穷和
A1	金额

(ii) · 3 marks

23 years.

$125000 - S_n = 125000(0.96)^n$。要求$(0.96)^n < 0.4$，得$n > \log 0.4 / \log 0.96 = 22.446\ldots$；至少23年。

M1	用无穷和减有限和
M1	对数不等式
A1	年数